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CSA1603 DWDN CLASS TEST

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1. Define data mining & list any two basic data mining tasks.

Definition;

data mining is the process of extracting useful, hidden, & meaningful patterns or knowledge from large amount of data.

Two basic data mining tasks

1. classification - assigns data into predefined classes.
2. cluster - groups similar data without predefined labels.

2. what is data cleaning?

data cleaning is the process of detecting & correcting errors, missing values, duplicate records,

& inconsistencies.

Two data quality issues;

- * missing value
- * noisy data

Handling missing values

1. Ignore tuple

eg;

Age	Salary
25	50000
?	45000

Remove the second record.

2. Fill missing values

Replace using

* Mean

* Median

* Mode

eg; Ages: 20, 25, 30, ?

$$\text{Mean} = (20 + 25 + 30) / 3$$

Handling noisy data

1. Binning

data: 10, 11, 12, 60, 13

After smoothing

10, 11, 12, 13, 13

2. Regression

Fit a regression line & replace abnormal values.

3. Clustering

reat distant object as outliers.

3. online data integration & list two challenges associated with it.

data integration combines data from multiple sources into one unified dataset.

Two challenges

- * Redundancy

- * Data inconsistency

Handling Redundancy

duplicate record are removed.

eg:

ID	name
101	John
101	John

keep only one record.

4. what is covariance analysis?

covariance measures how two variable change together.

- * positive covariance $\rightarrow$ variable increases together.

- * negative covariance $\rightarrow$ one increase while the

other datasets.

* zero covariance $\rightarrow$ no relationship.

Two covariance Formula

* population covariance

* sample covariance

5. Suppose two stock A & B have the following

stock A = 2, 3, 5, 4, 6

stock B = 5, 8, 10, 11, 14

Find whether they rise together or independently

X	Y	(X-4)	(Y-9.6)	product
2	5	-2	-4.6	9.2
3	8	-1	-1.6	1.6
5	10	1	0.4	0.4
4	11	0	1.4	0
6	14	2	4.4	8.8

sum = 20

sample covariance

both stock prices rise together (positively related),
not independent.

6. Explain the architecture of data mining system.

Components

Data sources : store raw data

Database : Improves data quality

Data cleaning module : Integrated data

Database server : Retrieve required data

Data Mining engine : perform mining algorithm

Pattern evaluation : select useful pattern

Knowledge Base : store domain knowledge

User Interface : display report, chart & graph

Interaction;

All modules work together to transform raw data into useful knowledge for decision making.

7. Find covariates

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

Mean

X	Y	$(X-6.2)$	$(Y-5.8)$	product
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56

Sum

Sample covariance

$$\text{Covariance} = -0.45$$

It indicates a weak negative relationship.

8. describe the major steps in data preprocessing steps;

- * data cleaning
- * data integration
- * data Transformation
- * data Reduction
- * data discretization

pipeline;

Raw data $\rightarrow$ cleaning $\rightarrow$ missing value handling
 $\rightarrow$ format standard $\leftarrow$ outlier detection
normalization $\rightarrow$ Feature selection $\rightarrow$ prepared dataset

Justification;

- * clearing unwanted errors.
- * Missing values are handled.
- * authors are detected.
- * Formats become consistent.
- * Normalization scale values.
- * Feature selection reduces complexity.

9. Perform equal-frequency binning

dataset:

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Total value = 27

choose 3 bins

each bin = 9 values

Bin 1

13, 15, 16, 16, 19, 20, 20, 21, 22

Mean = 18

smoothed

18, 18, 18, 18, 18, 18, 18, 18, 18

Bin 2

22, 25, 25, 25, 25, 30, 33, 33, 35

Mean ≈ 28

Smoothed

28, 28, 28, 28, 28, 28, 28, 28, 28

Bin 3

35, 35, 35, 36, 40, 45, 46, 52, 70

Mean ≈ 44

Smoothed

44, 44, 44, 44, 44, 44, 44, 44, 44

10. Explain data discretization & compare equal width & equal frequency binning.

data discretization converts continuous values into intervals.

Importance ;

- * Reduces noise
- * Simplifies analysis
- * Speeds up algorithm
- * Improves classification accuracy.

comparison;

Feature	equal-width	Equal-Frequency
Interval size	same width	same no of values
Data in bins	May differ	Equal count
Sensitive to out	yes	less sensitive
distribution	uneven	More balanced